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## **CS-8003(1)-CBGS**

### **B.E. VIII Semester**

Examination, December 2020

## **Choice Based Grading System (CBGS)**

### **Machine Learning**

*Time : Three Hours*

*Maximum Marks : 70*

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain K-Means algorithm with suitable example.  
b) Discuss in briefly about time series in ML.
2. a) Give detailed discussion on decision trees and boosting.  
b) Explain linear quadratic regulation.
3. a) Explain working principle of Independent components analysis.  
b) Give short notes on Real world ML.
4. a) Explain how back propagation algorithms helps in classification.  
b) Explain the steps in developing a machine learning algorithm.
5. a) What is the goal of support vector machine? How to compute the margin?  
b) Explain Bayes theorem.

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6.
  - a) Explain Hidden Markov model.
  - b) Discuss in brief elements of reinforcement learning.
7.
  - a) Explain different association rules with algorithms.
  - b) Explain principle component analysis with algorithm.
8. Write short notes on any two.
  - i) Big data and map reduce
  - ii) Common software for MC
  - iii) Subset selection

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